Recall: $\mathrm{Ext}^1(\Delta_\lambda, \nabla_\mu) = 0$

Prop: $\mathrm{Ext}^i(\Delta_\lambda, \nabla_\mu) = 0$

Proof: $0 \to K \to P_\lambda \twoheadrightarrow \Delta_\lambda \to 0$ Induct on i.

has stand. filtr.

$i=1$: known

$$0 = \mathrm{Ext}^i(P_\lambda, \nabla_\mu) \to \mathrm{Ext}^i(\Delta_\lambda, \nabla_\mu)$$

$$\overset{0}{\overset{\shortparallel}{\mathrm{Ext}^{i-1}(K, \nabla_\mu)}} \to 0$$

Claim: $\mathrm{Ext}^{i-1}(K, \nabla_\mu) = 0$ (stand. filt.)

Induct on length of stand. filt.

Know: $\mathrm{Ext}^{i-1}(\Delta_\lambda, \nabla_\mu) = 0$ (induction hyp.)

$0 \to K' \to K \twoheadrightarrow \Delta_\lambda \to 0$

$$\mathrm{Ext}^{i-1}(K', \nabla_\mu) \to \mathrm{Ext}^{i-1}(K, \nabla_\mu) \to 0$$

smaller length

$\overset{\shortparallel}{0}$ $\qquad$ $\overset{\shortparallel}{0}$

Exercise If M standard filt, N costand. filt. $\Rightarrow$ $\mathrm{Ext}^i(M, N) = 0$ $\forall i > 0$.

Def: $M \in \mathcal{O}$ is tilting if it has both standard filtration and costandard filtration.

$\Rightarrow$ $\mathrm{Ext}^i(T_1, T_2) = 0$ $\qquad$ Ex. $T = \bigoplus T_i$ "generating" tilting cty.

tilting

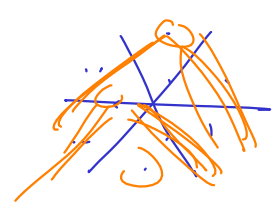
Keller, Derived cat & tilting

Rickard, Morita theory for derived cat

Great tilting theory (derived Morita theory) $\qquad D(\mathcal{O}_\lambda) \simeq D(\mathrm{Hom}(T, T))$

Thm (BGG reciprocity) $\qquad (\Delta_\mu : M) =$ multiplicity of M standard filtration

$= [L_\lambda : N] =$ mult. of L_λ in JH filtration



$$(\Delta_\mu : P_\lambda) = [L_\lambda : \nabla_\mu] = [L_\lambda : \Delta_\mu] = (\nabla_\mu : I_\lambda)$$

Proof: Claim $= \dim(\mathrm{Hom}(P_\lambda, \nabla_\mu))$

① $(\Delta_\mu : P_\lambda) = \dim(\mathrm{Hom}(P_\lambda, \nabla_\mu))$ — induct on standard filtration of P_λ

$\boxed{\cdots \subseteq M' \subseteq M \subseteq \cdots \subseteq P_\lambda}$

$0 \to M' \to M \to \Delta_\nu \to 0$

$0 \to \underset{\delta_{\nu\mu}}{\mathrm{Hom}(\Delta_\nu, \nabla_\mu)} \to \mathrm{Hom}(M, \nabla_\mu) \to \mathrm{Hom}(M', \nabla_\mu)$

$\mathrm{Ext}^1(\Delta_\nu, \nabla_\mu) = 0$

dim of Hom increases if $\nu = \mu$

$(\Delta_\mu, -)$ " " " "

So, $=$.

induct in comp. series for ∇_μ.

② $[L_\lambda : \nabla_\mu] = \dim(\mathrm{Hom}(P_\lambda, \nabla_\mu))$

$0 \to M' \to M \to L_i \to 0$

$0 \to \mathrm{Hom}(P_\lambda, M') \to \mathrm{Hom}(P_\lambda, M) \to \underbrace{\mathrm{Hom}(P_\lambda, L_i)}_{\dim = \delta_{\lambda\nu}} \to 0$.

indecomposable iff $\lambda = \nu$

$P_\lambda \twoheadrightarrow L_\nu$ $\neq 0$ or surj.

$\oplus \cup$

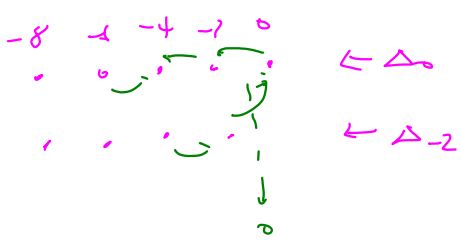
P_ν

Same inverse induction

Ex: $(\Delta_\mu : P_\lambda) = [L_\lambda : \Delta_\mu]$

P_{-2} = big projective

multiplicity 1
$$\begin{cases} (P_{-2} : \Delta_{-2}) = [L_{-2} : \Delta_{-2}] = 1 \\ (P_{-2} : \Delta_0) = [L_{-2} : \Delta_0] = 1 \end{cases}$$



P_λ multiplicity 1

antidominant

$\boxed{(\Delta_\mu : P_\lambda) = 1 \quad \forall \mu}$

$[L_\lambda : \Delta] = 1$?

$\mathrm{Hom}(\Delta_\lambda, \Delta) =$

Translation functors λ_2, λ_1 weights of $\mathfrak{g}$ Assume: $\lambda_2 - \lambda_1$ integral, regular

μ^+ be the linked dominant weight to $\lambda_2 - \lambda_1$

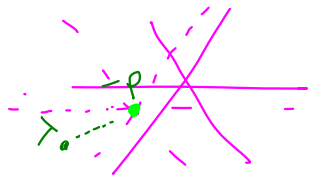
$T_{\lambda_2\lambda_1} : \mathcal{O}_{[\lambda_1]} \hookleftarrow \mathcal{O} \xrightarrow{- \otimes L} \mathcal{O} \to \mathcal{O}_{[\lambda_2]}$ ← fd. exact

Rep. T exact. $\Rightarrow$ take proj to proj.
inj to inj.

Prop. λ antidominant, integral. $P_\lambda \simeq I_\lambda$, i.e. self dual. $\Rightarrow$ tilting.

$P_\lambda \simeq T_{\lambda,-\rho} L_{-\rho}$ $\quad (L_{-\rho} \cong \Delta_{-\rho} \simeq \nabla_{-\rho} \simeq P_{-\rho} \simeq I_{-\rho})$

Proof $\Xi = T_{\lambda,-\rho} P_{-\rho} \overset{①}{\twoheadrightarrow} P_\lambda$

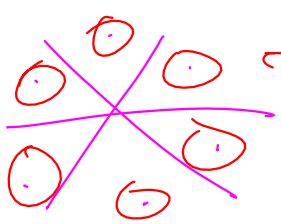


① There is a nonzero map $\Xi \longrightarrow L_\lambda$ $\Rightarrow$ P_λ is a summand of Ξ projective

$\tilde{\lambda}$ dominant linked to λ Ξ = summand (λ-linkage) of $\Delta_{-\rho} \otimes L_{-\lambda-\rho}$

filtration w/ quotient given by
$-\rho + (\cancel{\text{lowest wt of } L_{-\lambda-\rho}})$
$\cancel{= -\rho + w_0 \cdot (-\lambda-\rho)}$
$\cancel{= -(\rho + w_0 \cdot \lambda + \rho)}$

② count multiplicity of simples
$= 1$.



$\leftarrow$ show μ is linkage down for λ.

L_μ

$\mu = \lambda - \rho \quad \pm 1$.

$\Longrightarrow \quad \Xi \simeq P_\lambda \simeq I_\lambda$ self dual.